

Cinemática y movimiento.

Introduction

Kinematics is the branch of physics that studies **how objects move**, without worrying about why they move.

Example: when you ride a bicycle, you can pay attention to **where you are, what path you follow, how fast you are going, or whether you are accelerating**. All of that is studied by kinematics.

Motion is relative

What does this mean? A person sitting next to you on the bus is at rest from your point of view. But from the perspective of someone sitting at the bus stop, that person is moving.

! Very important

Motion always depends on the observer: something moves when its position changes with respect to a point that we choose (in this case, relative to the observer). This point is called the **reference point** or **origin of the reference system**.



Figure 1: Bus stop

Reference system and position

Imagine you are talking to a friend so they can find you in a park, and you tell them you are one hundred meters from the entrance. With that information, you have given a point that acts as the origin of the reference system ("the entrance") and the position ("one hundred meters from that entrance").

! Very important

- **Reference system:** a set of axes and a point, which is the origin, from which we measure positions.
- **Position (x):** the place where an object is with respect to the origin of the reference system. Unit: meter (m).

Trajectory

Trajectory: the path followed by an object, that is, the set of its positions as time passes.

Distance traveled and displacement

! Very important

Distance traveled (s): the total distance covered by a body during its motion. Unit: **meter (m)**.

Displacement (Δx):

$$\Delta x = x_f - x_i$$

Speed and velocity

- **Speed:** how fast something moves. Unit: **m/s**.
- **Velocity:** speed + direction.

! Very important

$$v = \frac{s}{t}$$

$$v = \frac{s}{t} \rightarrow v \cdot t = s$$

$$v = \frac{s}{t} \rightarrow t = \frac{s}{v}$$